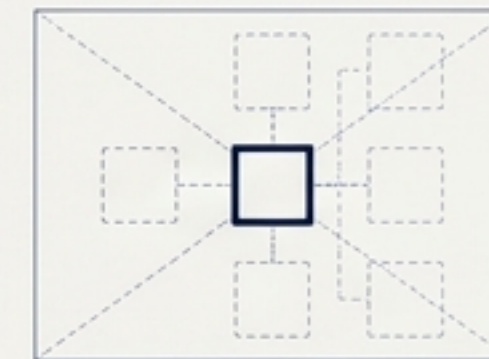
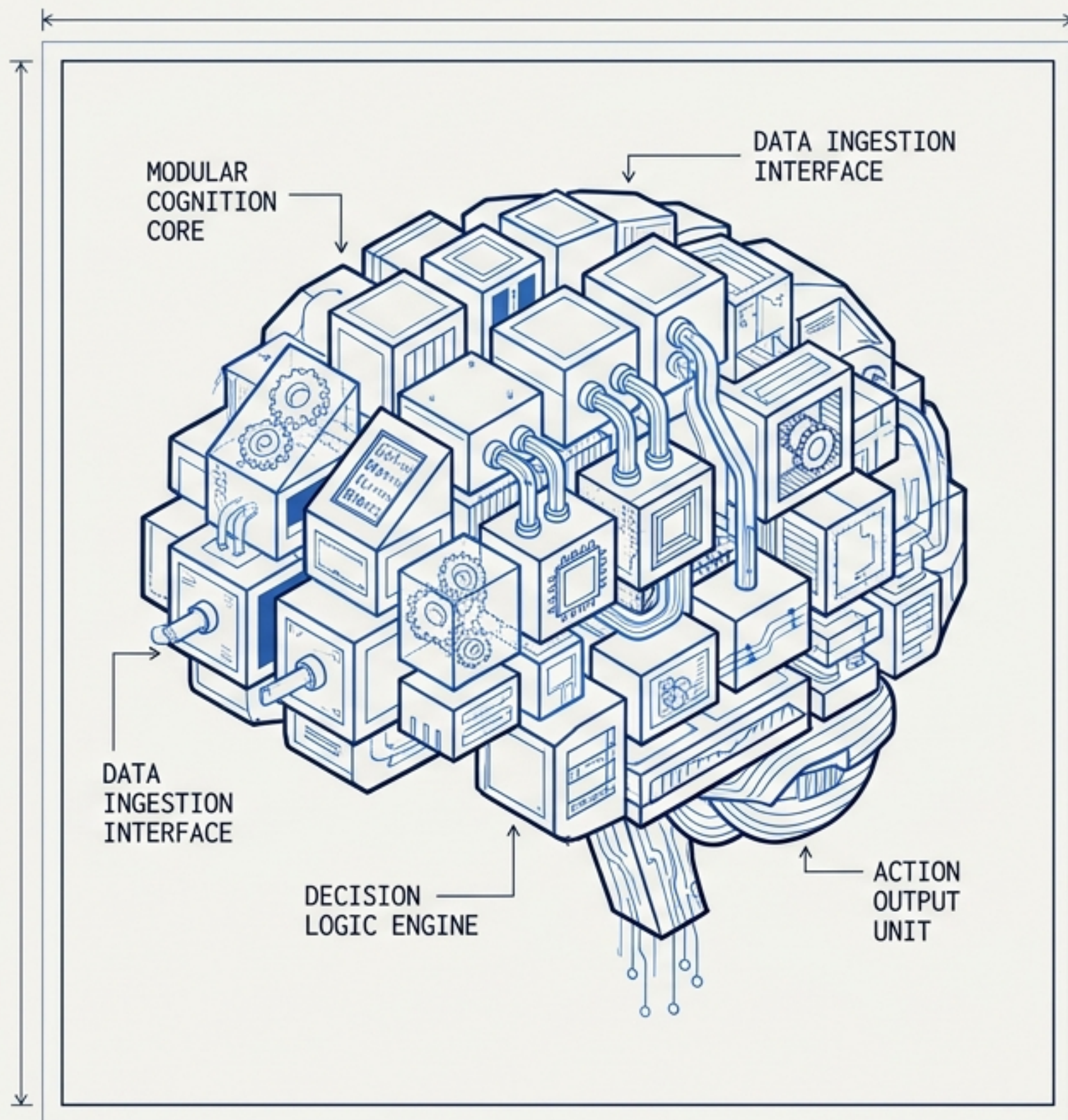




BUILDING AI AGENTS: A COMPREHENSIVE GUIDE

From passive text prediction to autonomous systems

EXTRAORDINARY REASONING

Language models possess advanced logic, but are historically confined to static training data.

LOGIC CORE V2.0

BRIDGING THE GAP

Agents evolve from passive text predictors into active entities.

EVOLUTION PATHWAY → ACTIVE_STATE

REAL-WORLD INTERACTION

Systems are equipped to fetch live data, execute code, and write to databases.

FUNCTION(FETCH_DATA, EXECUTE_CODE, WRITE_DB)

THE BLUEPRINT

Mastering tools, memory, state management, and evaluation.

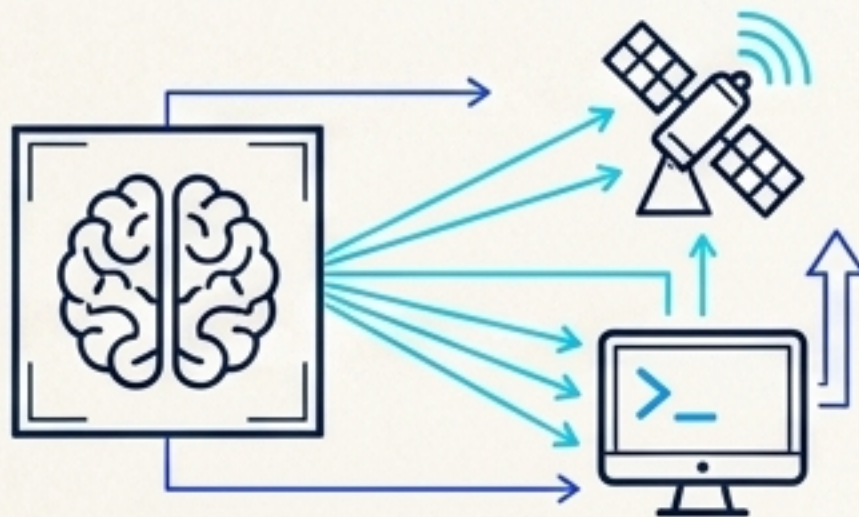
ARCHITECTURE: TOOLS | MEMORY | STATE | EVAL

The Power & Anatomy of Tools

THE POWER OF EQUIPPED AGENTS

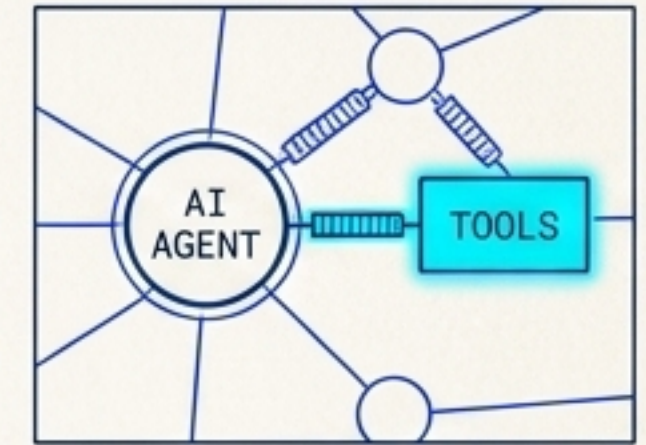


Locked-in Analyst: Risks hallucinations or stale facts.



Equipped Agent: Fetches live data and remains factually grounded.

ANATOMY OF A TOOL



Name

A unique, descriptive identifier the model uses to reference the tool.

Description

A natural-language explanation of what the tool does. The model reads this to decide when to call the tool.

Parameters

A JSON Schema object describing every argument.

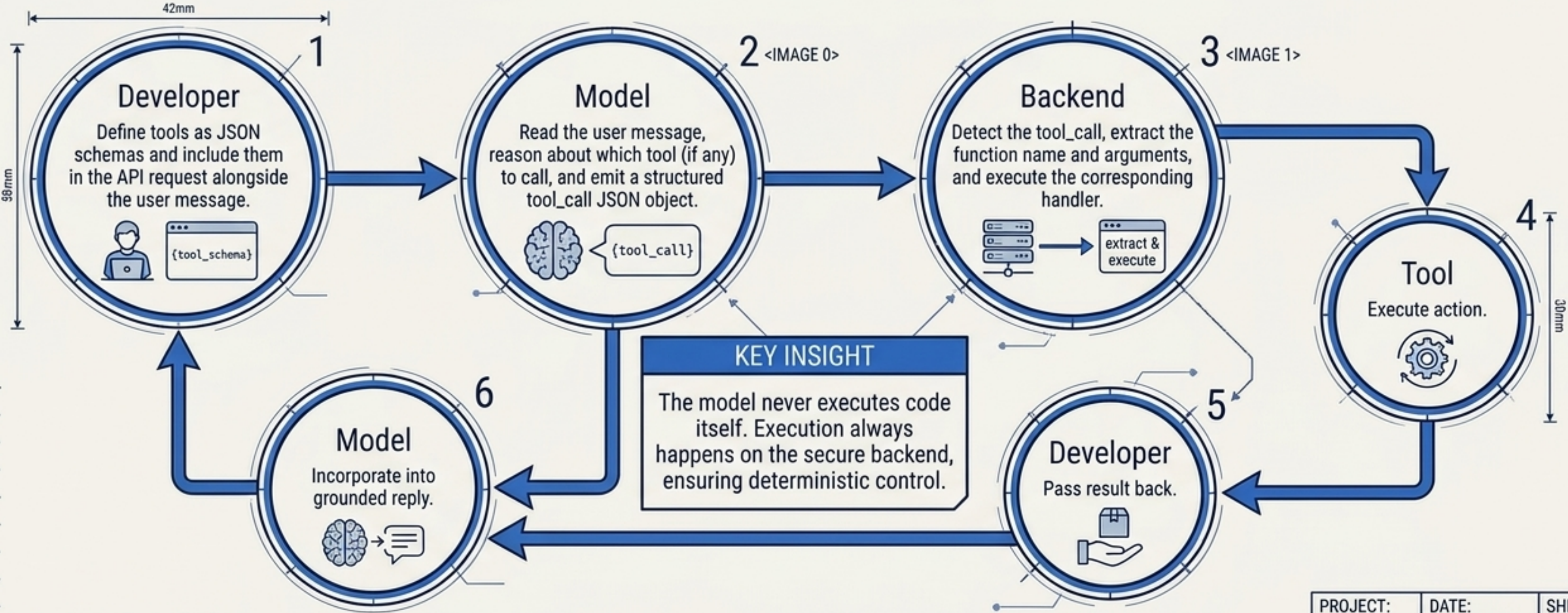
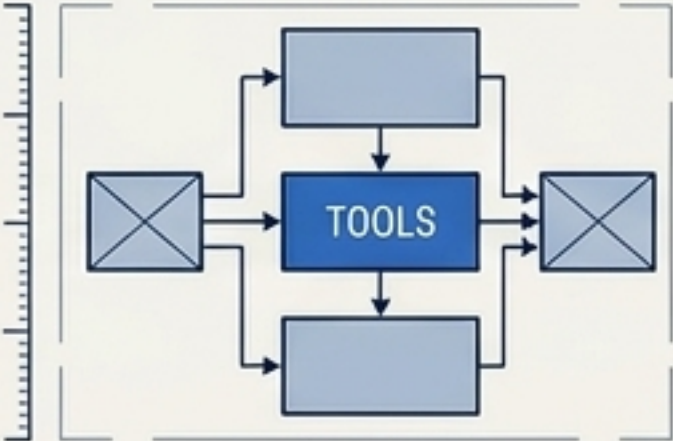
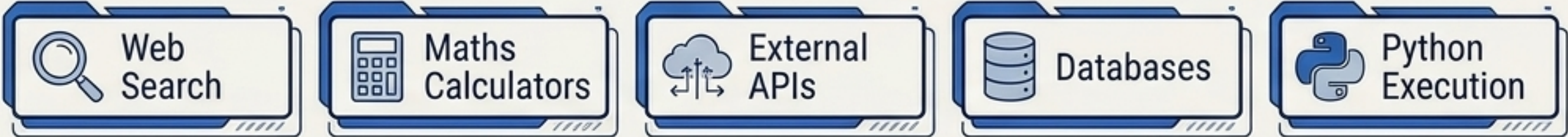
```
{  
  "type": "object",  
  "properties": {  
    ...  
  }  
}
```

Handler

The underlying Python function that executes the action.

```
def tool_handler(args):  
    # implementation
```

Tool Categories & The Execution Lifecycle



Structured Outputs & Pydantic Validation

Unstructured Prose

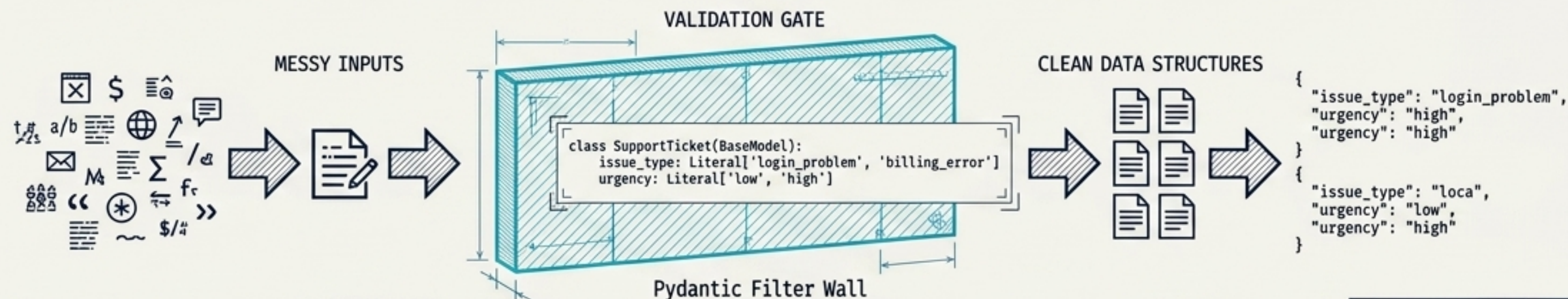
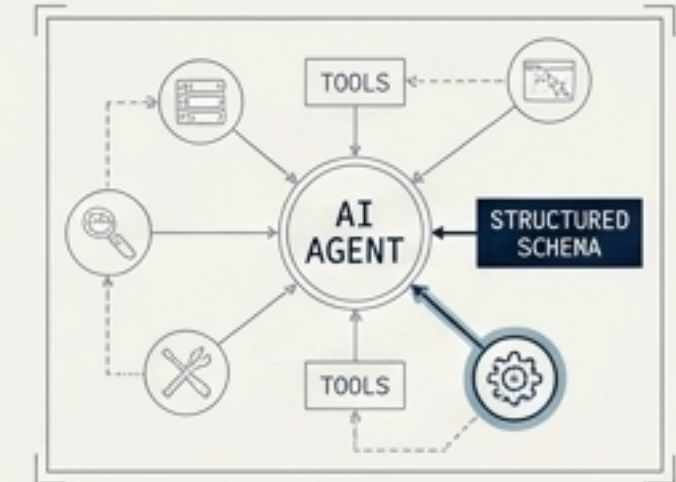
maybe... could be... use it...
 difference...
 I think the issue is...
 can't want to
 please fix it.

Reliable systems cannot parse creative prose.
Without rigid formatting, downstream functions break.

Machine-Readable JSON

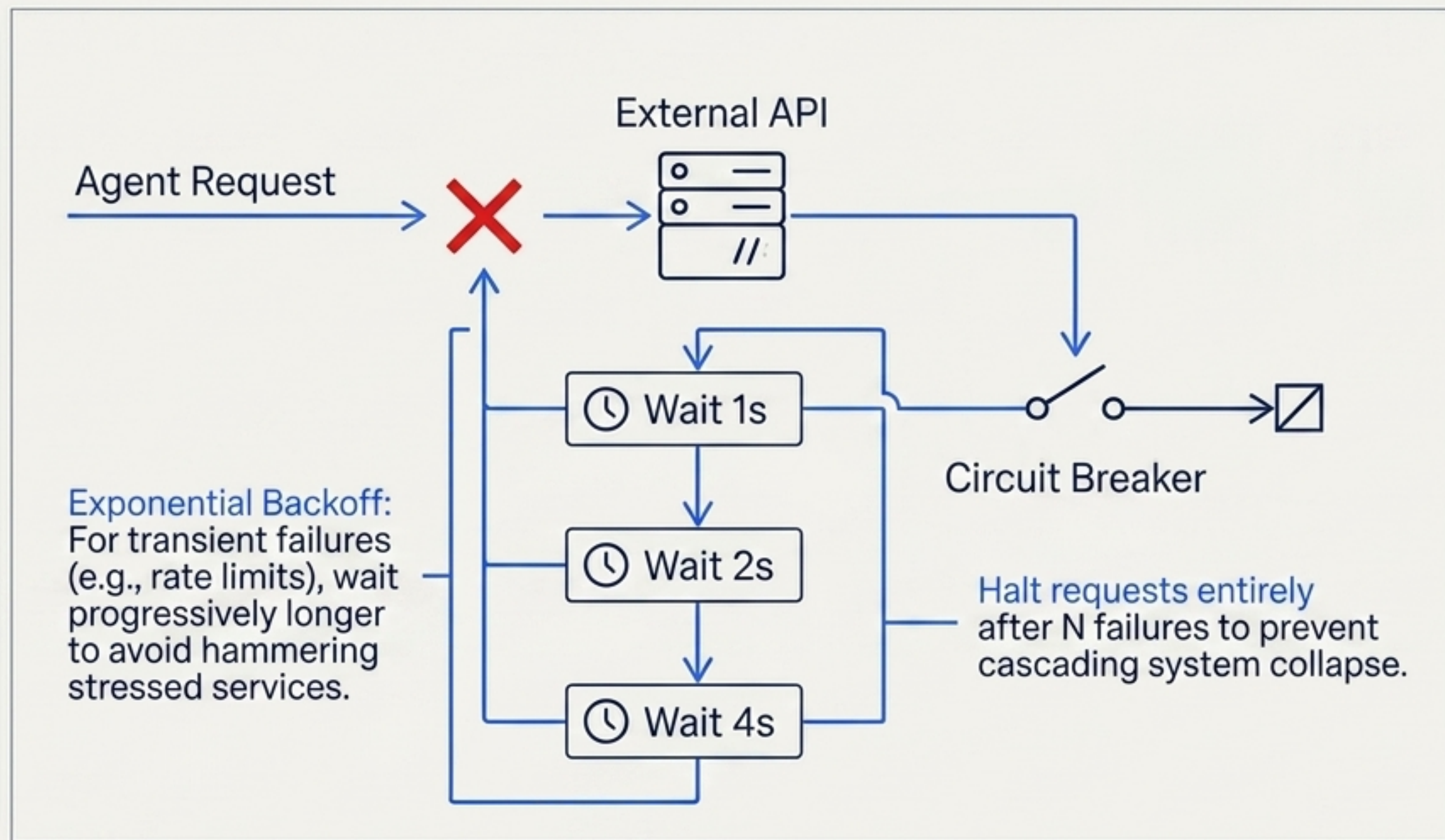
```
{
  "type": "SupportTicket",
  "data": {
    "issue_type": "login_problem",
    "urgency": "high"
  },
}
```

Modern systems embed schemas at the API level, enforcing strict boundaries before execution begins.

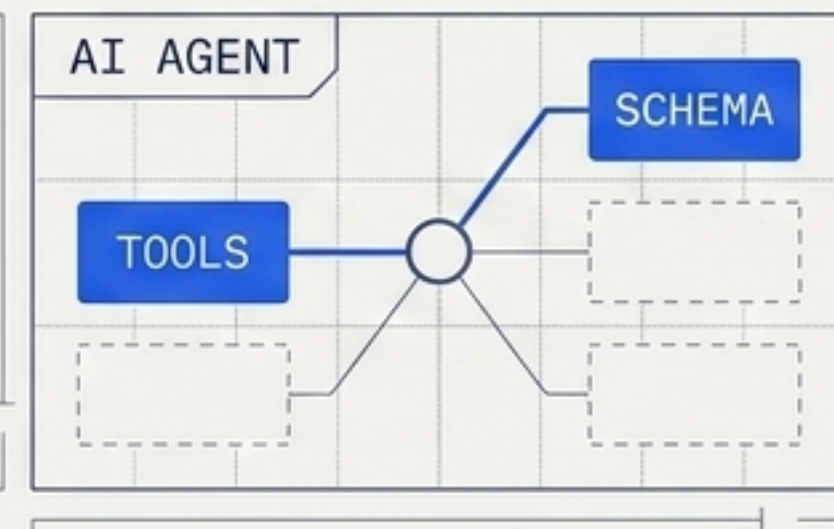


Data structures support strict enums and literals to completely prevent hallucinated variables.

[illegible]



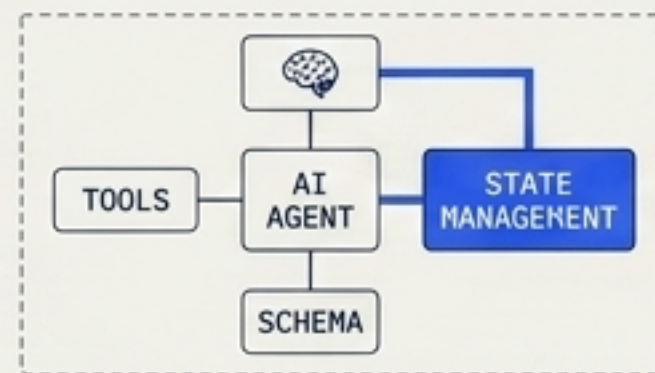
MINI-MAP



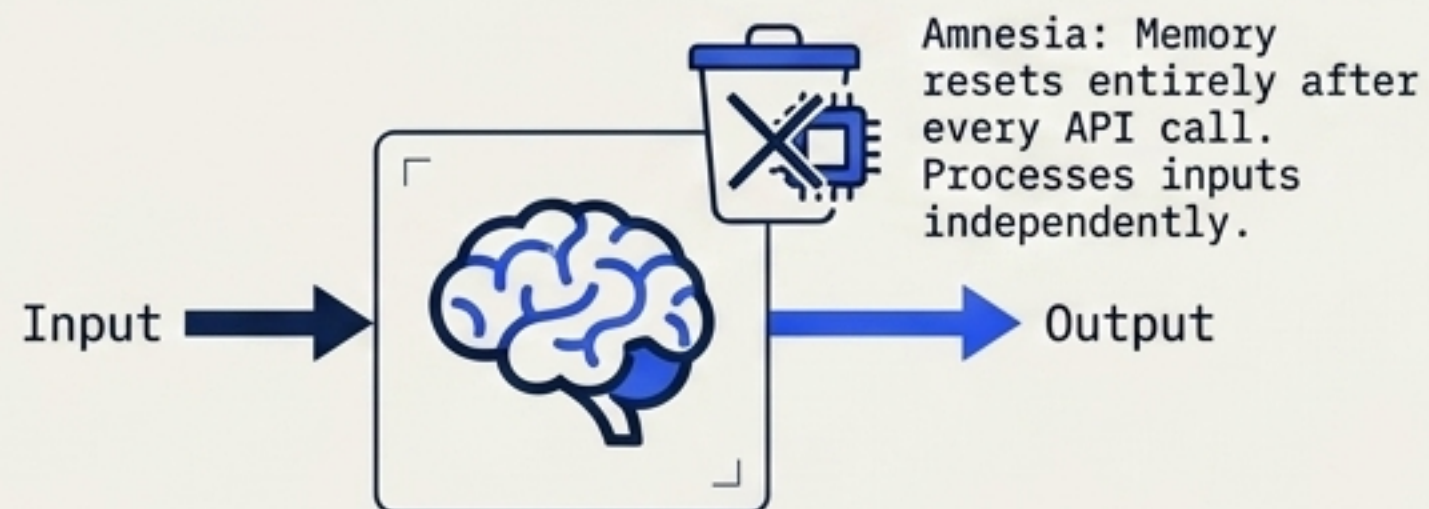
Designing Resilient Agent Systems

Never allow unhandled exceptions to reach the model directly.
Always establish fallbacks to secondary providers or cached data.

Agent State Management

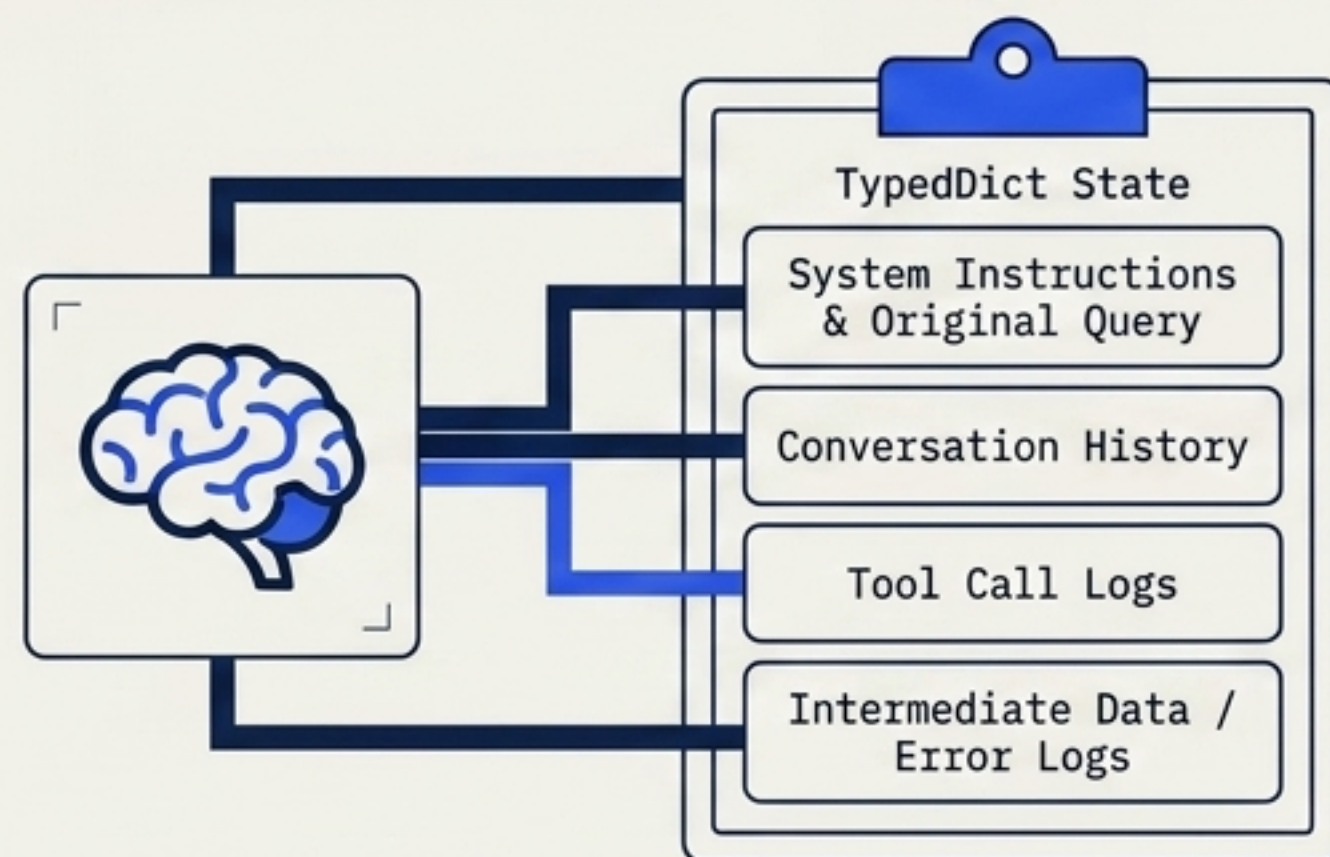


Stateless LLM



Without state, the agent treats every interaction as a new, isolated event. Context and progress are lost after each turn.

Stateful Agent

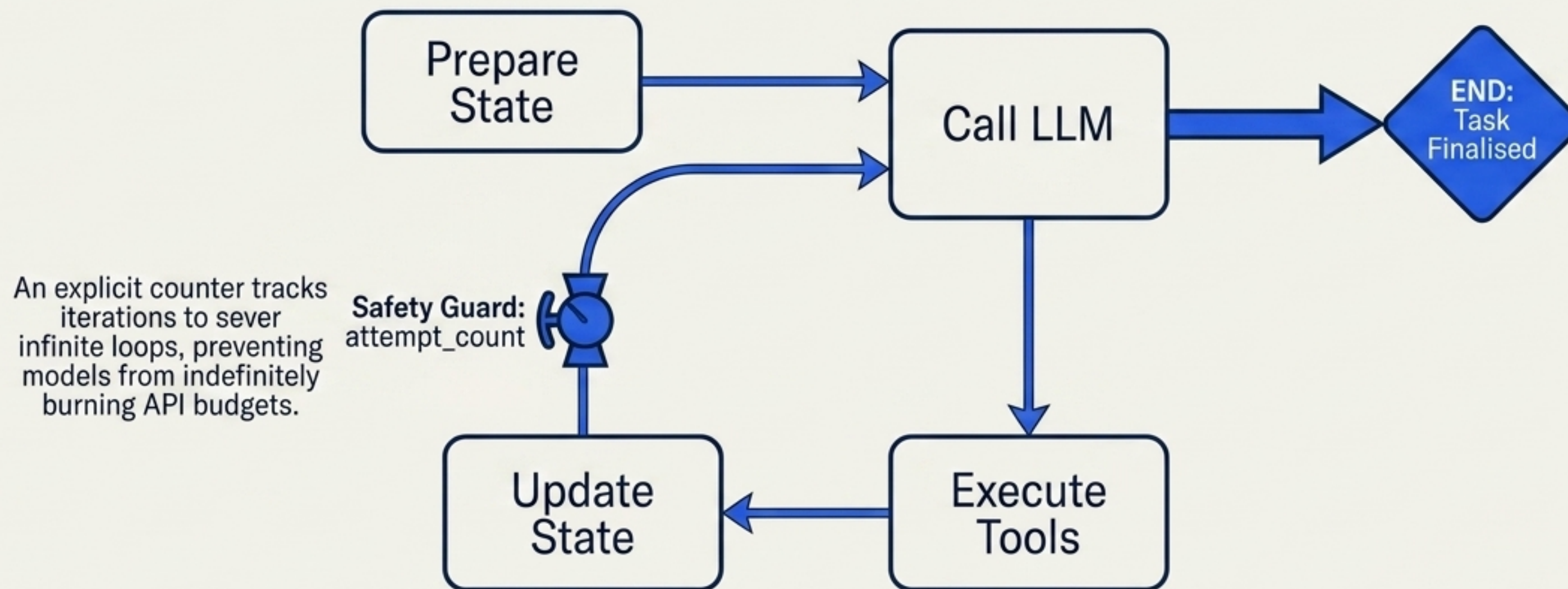
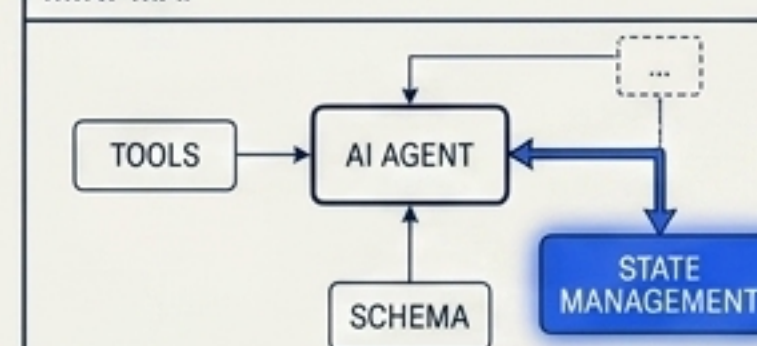


Maintains a persistent scratchpad of the ongoing task. State allows agents to recover from errors and track when a task is finalised.

Modelling Agents as State Machines

Routing execution systematically between finite, predictable steps.

MINI-MAP



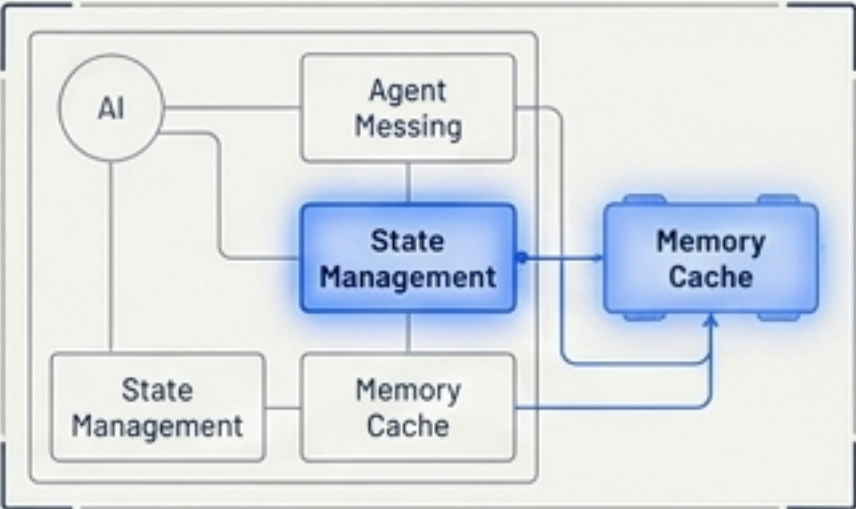
SCHEMATIC TITLE:
Agent State Management

DATE:

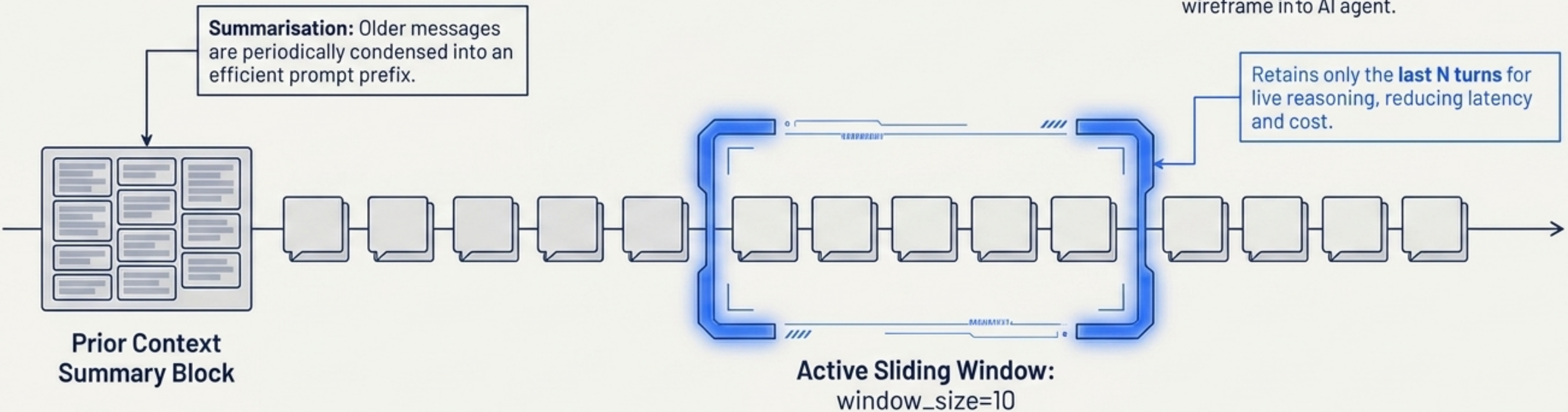
YYYY-MM-DD

Short-Term Memory Strategies

Managing conversation history within strict token limits.



A **headline map** is a persistent mini-map wireframe into AI agent.



This strategy bounds token costs while preserving full underlying history for logging and audits.

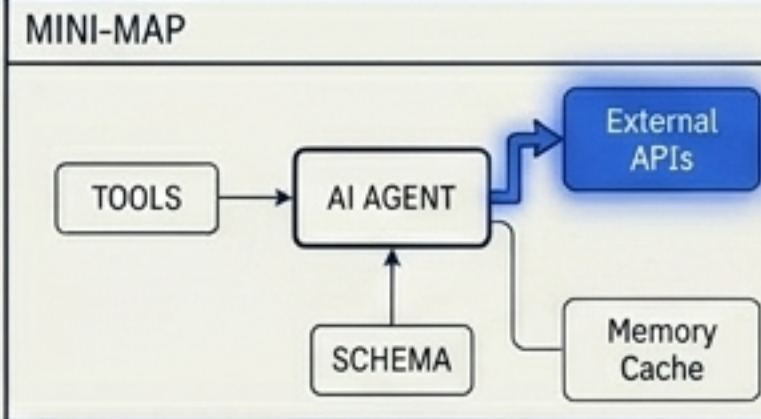
SCHEMATIC TITLE:
Short-Term Memory Strategies

DATE

YYYY-MM-DD

External APIs & Authentication

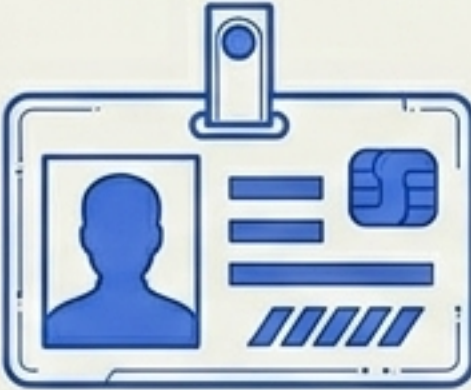
The universal protocols for interacting with the web.



GET	Read	Fetch data without changing anything.
POST	Create	Execute actions (booking flights, sending messages).
PUT	Update	Modify existing data.
DELETE	Remove	Delete records (use with caution).



API Keys: Static secrets granting app-level access. Never expose these in code.



OAuth 2.0: Required for user-specific delegated access (e.g., reading a user's personal calendar).

SCHEMATIC TITLE:
External APIs & Authentication

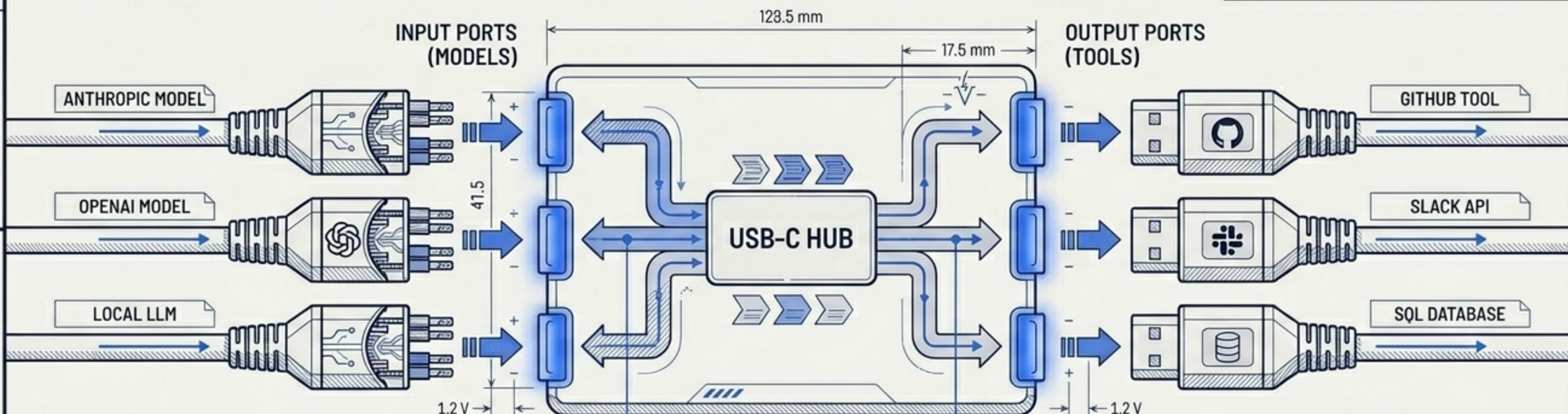
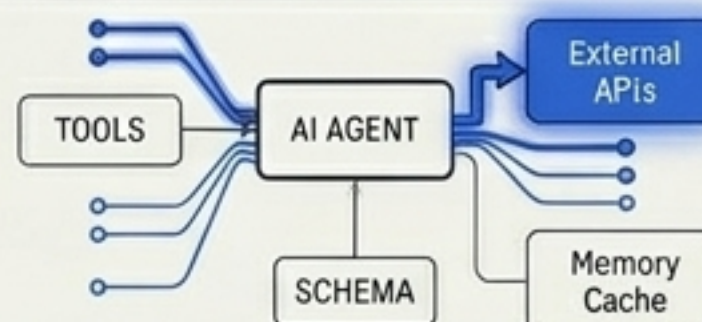
DATE:

YYYY-MM-DD

Model Context Protocol (MCP)

The open standard for universal tool connectivity.

MINI-MAP



The 'USB-C' of AI:

MCP replaces custom-coded handlers with universal, plug-and-play tool servers. Standardized connectivity for seamless integration across models and tools.

Intent Layer:

Models standardize exactly how they ask to use a specific tool. Defines tool call structure, parameters, and expected output format in a unified schema.

Execution Layer:

The MCP server handles all authentication, retries, and API execution securely. Abstracts away tool-specific implementation details from the core model logic.

SCHEMATIC TITLE:

Model Context Protocol (MCP)

DATE:

YYYY-MM-DD

WEB SEARCH AGENTS & GROUNDING

Fighting hallucination by equipping models with a browser.



MINI-MAP

WEB SEARCH

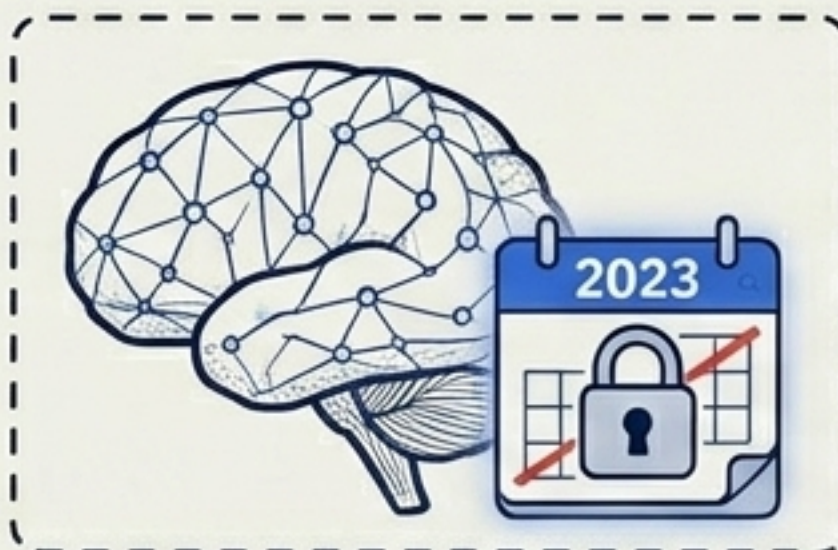
AI AGENT

External APIs

SCHEMA

Memory Cache

WITHOUT SEARCH

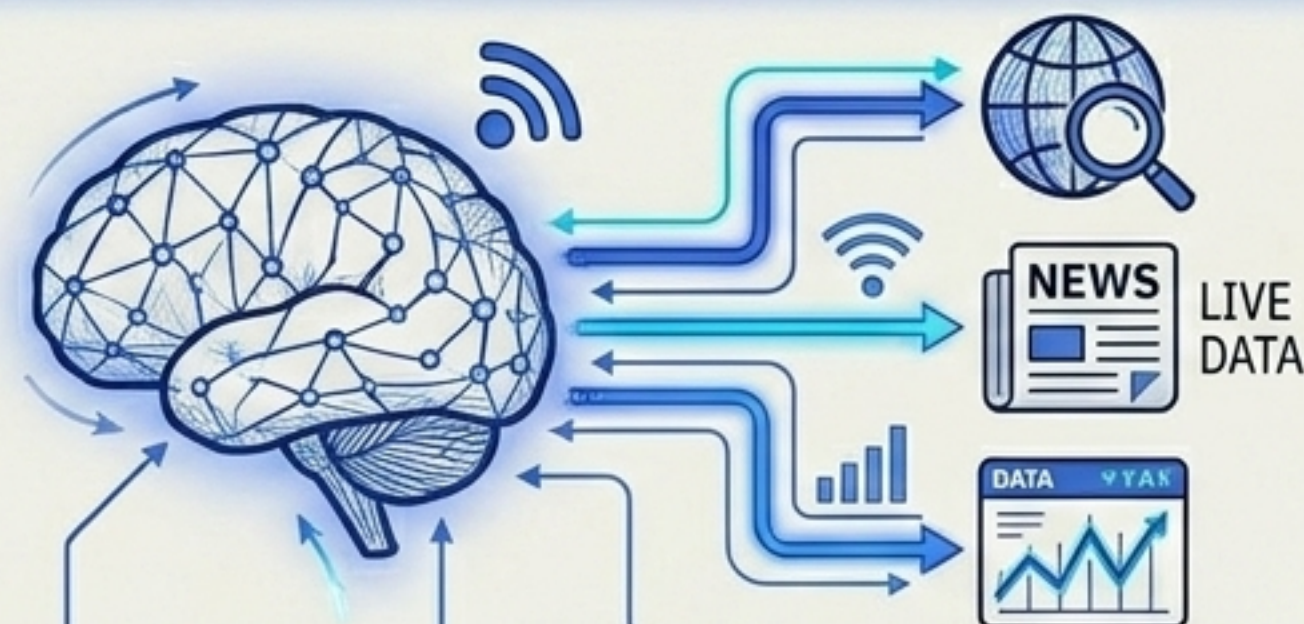


Knowledge is frozen at training cutoff.

High risk of confidently stating stale facts.

Produces unverified text and plausible-sounding hallucinations.

WITH SEARCH



Retrieves live information at query time.

Accesses current market data, news, and live scores.

Grounding: Forces the model to cite retrieved URLs, creating auditable outputs.

NOTE: Tools like Tavily return clean, LLM-optimised text rather than messy, token-heavy HTML.

SCHEMATIC TITLE:

Web Search Agents & Grounding

DATE:

YYYY-MM-DD

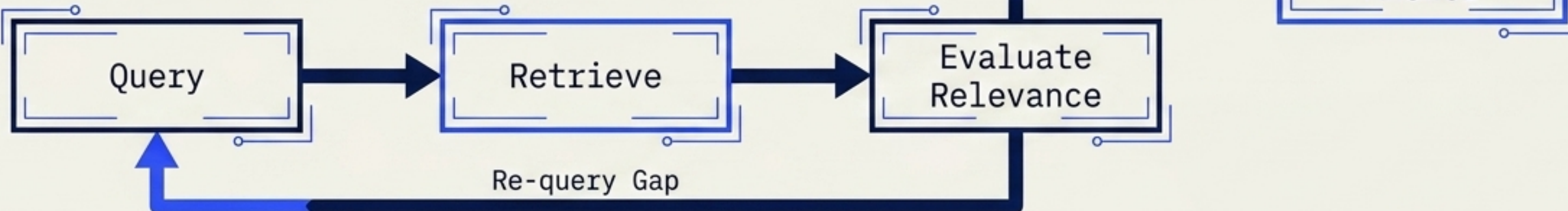
Active Reasoning: Agentic RAG

Passive RAG



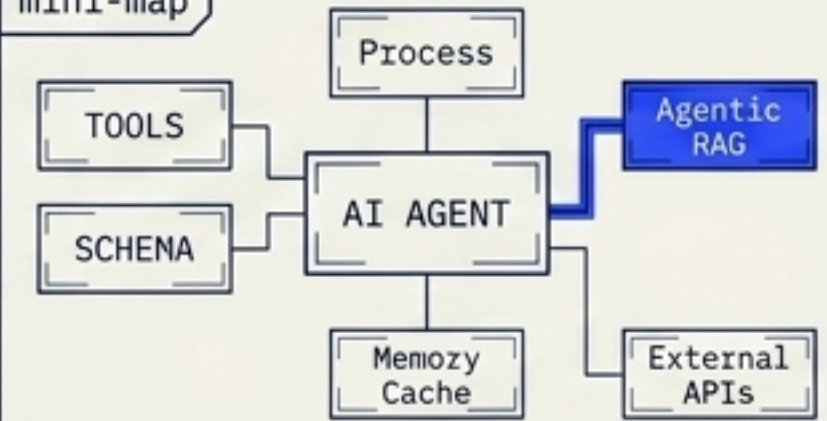
Retrieves once with the original query.
Generates an answer even if retrieved data is irrelevant.

Agentic RAG



Evaluates relevance before using content.
Identifies gaps and re-queries rather than hallucinating.

mini-map



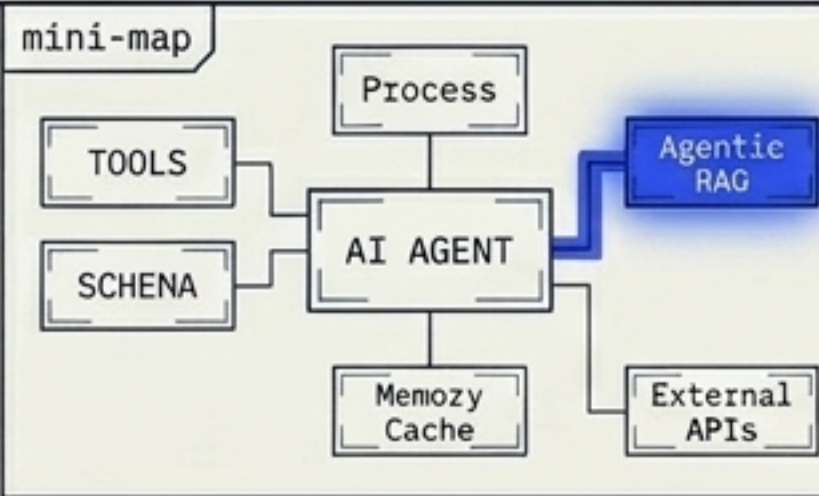
SCHEMATIC TITLE:

Agentic RAG

DATE: YYYY-MM-DD

QUERY REFORMULATION STRATEGIES

The cognitive toolkit an agent uses when a search hits a dead end.



FAILED INITIAL QUERY:
"Zillow iBuying failure"

DECOMPOSITION
(BREAKING DOWN)

ABSTRACTION
(MOVING UPWARDS)

What algorithmic assumptions
did the model make?

How did 2021 market
conditions differ?

↳ Breaking a complex question into
multiple distinct sub-questions.

iBuying business
model risks

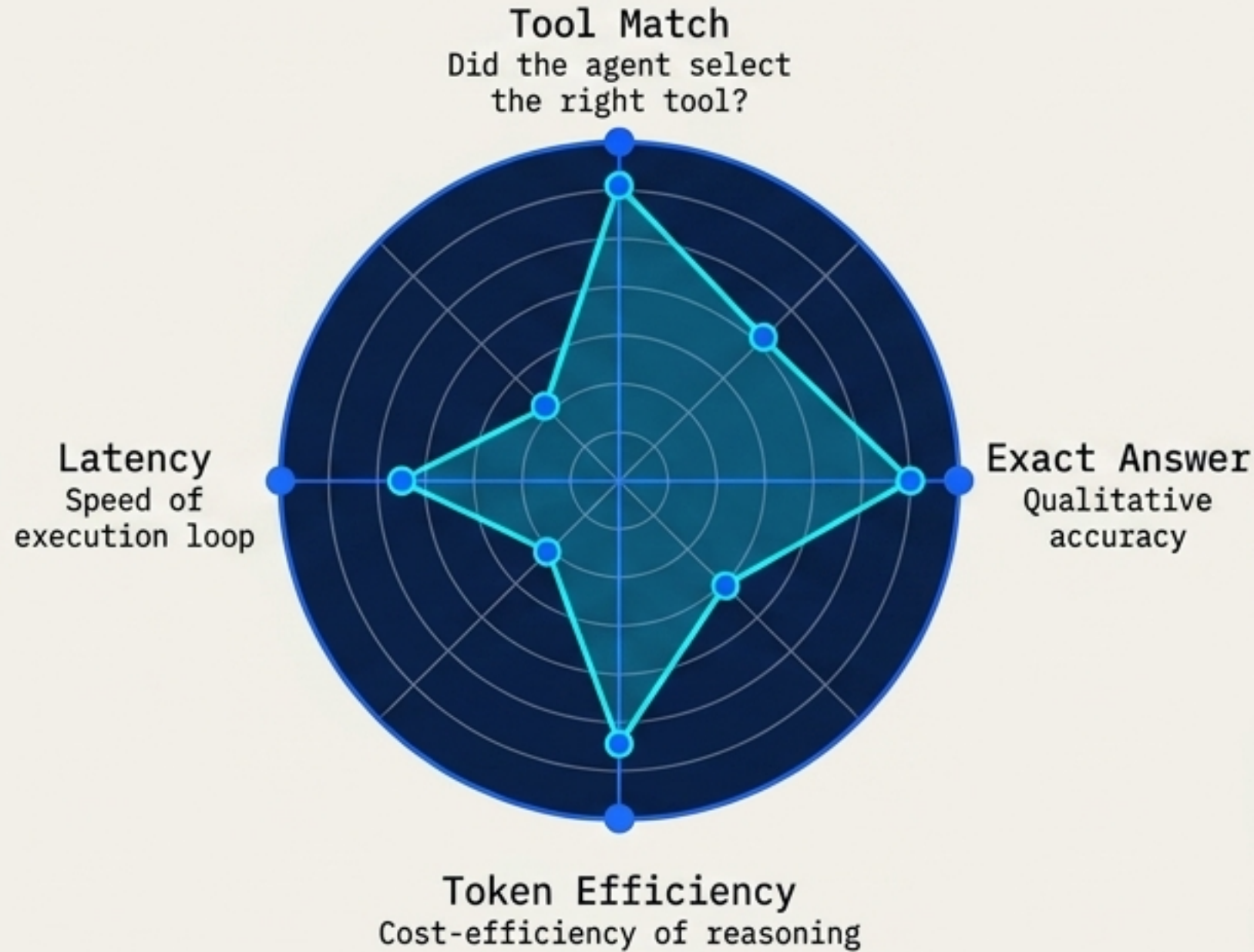
↳ Moving from hyper-specific queries
to broader thematic searches.

Effective reformulation is the core skill that makes agentic search superior to simple vector lookups.

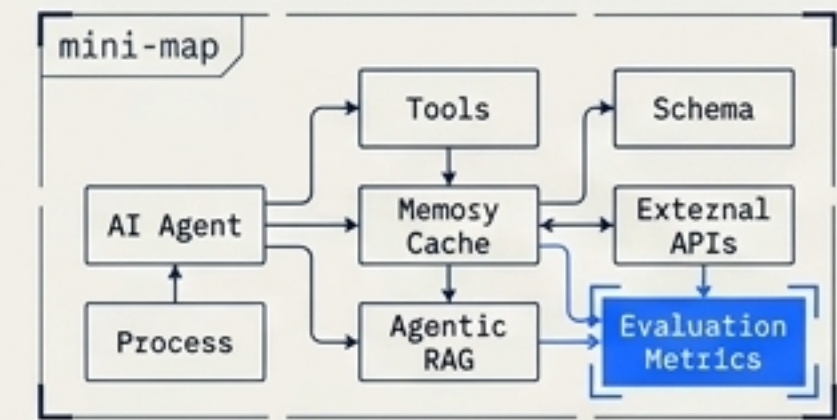
SCHEMATIC TITLE:
Query Reformulation Strategies
DATE: YYYY-MM-DD

Evaluating Agent Performance

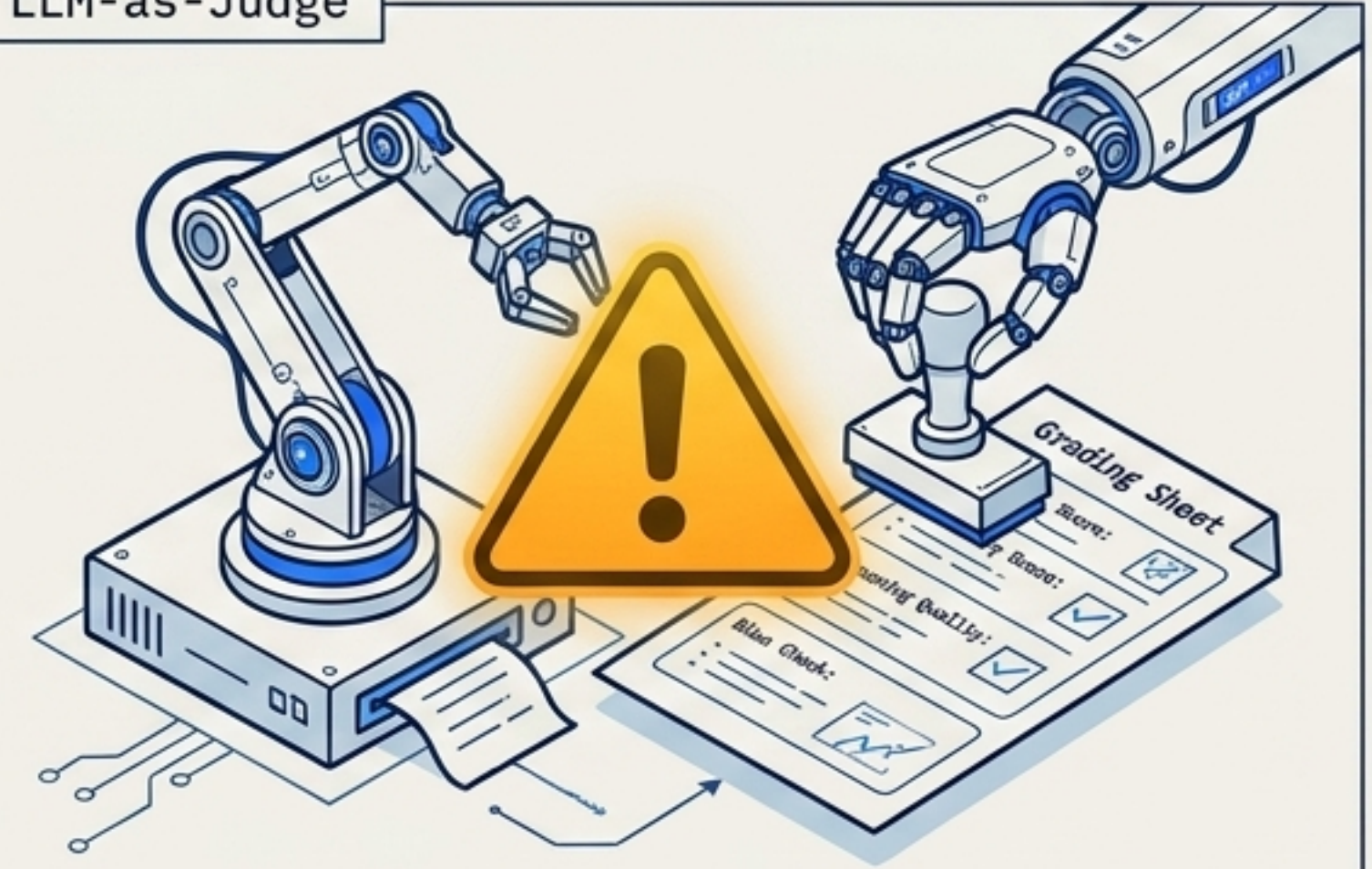
Performance Radar



Non-deterministic systems require tracking full execution trajectories, not just the final text.



LLM-as-Judge



Powerful for scoring qualitative answers, but requires strict guidelines to mitigate built-in biases.

SCHEMATIC TITLE:
Evaluating Agent Performance

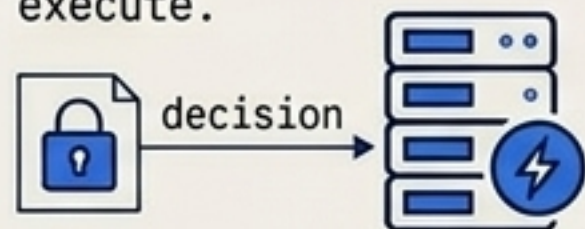
DATE: YYYY-MM-DD

KEY TAKEAWAYS FOR PRODUCTION AGENTS

Agents are no longer black boxes—they are composable, engineered systems.

KEEP CONTROL BACKEND

Models decide; your secure servers execute.



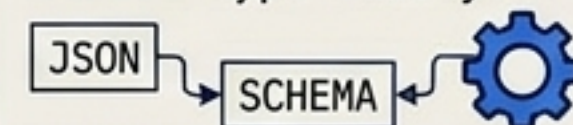
BUILD RESILIENCE

Assume APIs will fail; code robust retries and circuit breakers.



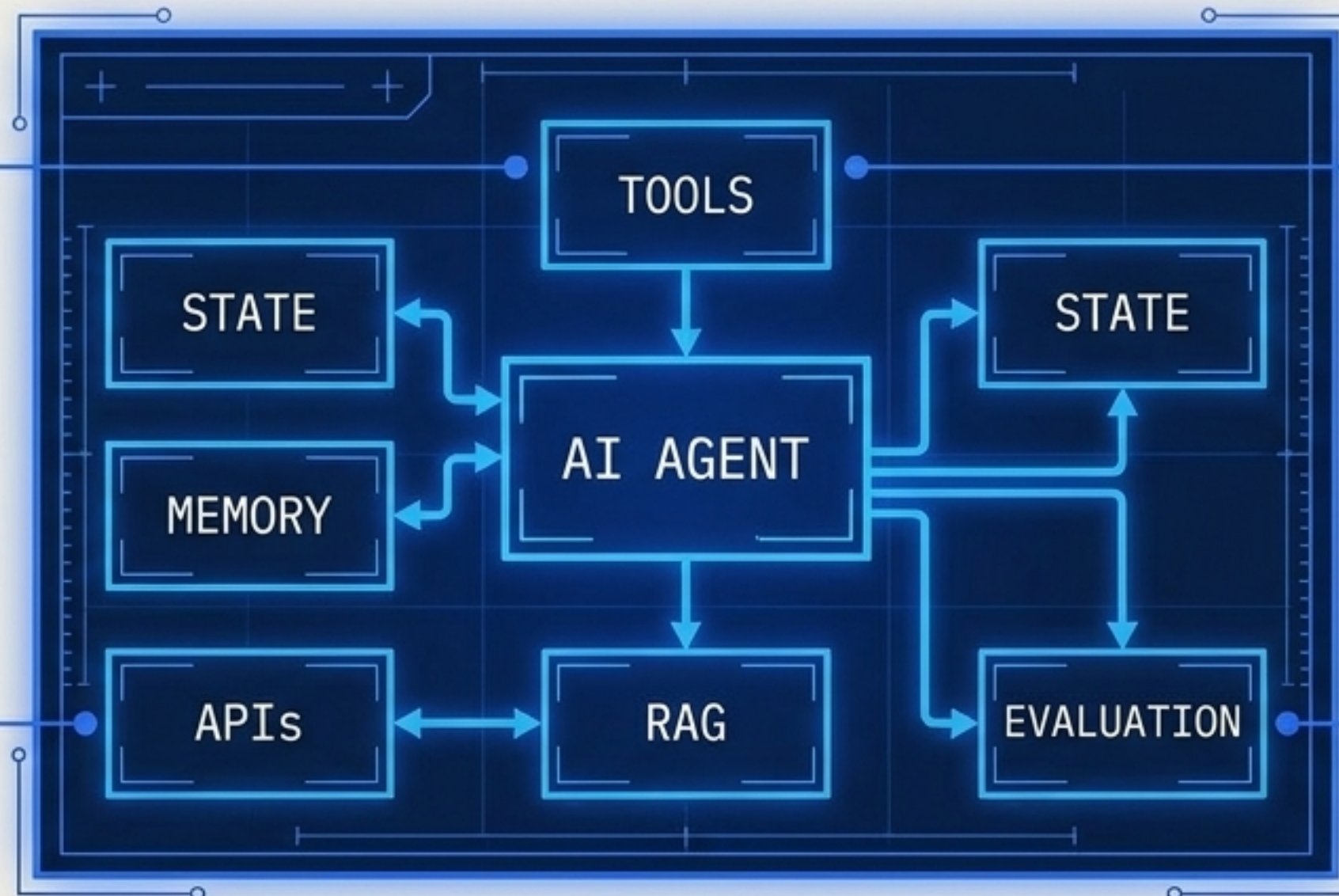
ENFORCE STRUCTURE

Use Pydantic and JSON schemas to mandate strict type safety.



MANAGE STATE EXPLICITLY

Bound token limits with sliding windows and cap execution loops.



SCHEMATIC TITLE:

Key Takeaways for Production Agents

DATE: YYYY-MM-DD